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Bosch-Norm

BOSCH

N28 AS0715 S004

2019-05-06

Free-cutting steel

With limited chemical composition

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Area of validity, binding force, freedom to disclose	
Valid for	Bosch Group
Binding for	Bosch Group with the exception of BEGE and the areas special mechanical engineering business, as well as single-item and prototype production.
Binding due to	Guideline RB/GF 131
Recommended for	BEGE and the areas special mechanical engineering business, as well as single-item and prototype production.
Disclosure	Restricted in accordance with the Guideline RB/GF 108 - Any disclosure to third parties requires the consent of the supplier named in this Bosch-Norm.

This Bosch-Norm is binding for suppliers if it is part of the order.

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General information	
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In case of doubt, the original language edition of this Bosch-Norm applies.
The comma is used as the decimal marker.
Valid edition only in NormMaster. No change service for copies.

Standard quality

1 Scope

This Bosch-Norm specifies the technical terms of delivery and additional requirements for the free-cutting steel 11SMn30 with a restricted chemical composition. In addition to the data in this Bosch-Norm, all requirements of the external standard have to be met.
Typical use: for general use

Note: Structure of the Bosch designation, see N28 AS (internal document of the Bosch Group)

2 Material designation

Quality standard	DIN EN 10277:2018-09
Material abbreviation	11SMn30
Material number	1.0715

3 Qualities, forms of delivery, dimensional deviations and requirements

The sections 8, 9 and 10 are valid with regard to the specific material properties. The sections not listed apply to all qualities. In the column “Requirements per section”, these specific requirements are allocated to the qualities.

Bosch designation	Bosch material part number	Form(s) of delivery	Dimensional deviations	Requirements per section
AS0715-C01Z-SR-B4-2E01	5 107 508 1..	round bars	N23A 2E01	8.2; 9.1; 10.1
AS0715-A01Z-SR-B3-2E01	5 107 508 2..	round bars	N23A 2E01	8.1; 9.2; 10.4
AS0715-A02Z-SR-B3-2E01	5 107 508 3..	round bars	N23A 2E01	8.1; 9.2; 10.3.2; 10.4

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4 Manufacturing

The steelmaking processes (e.g., melting, secondary metallurgy, casting) are to be ensured with regard to high requirements in order to avoid macroscopic defects, non-metallic inclusions and defects.
The steelmaking and production of the semi-finished material up to the product to be delivered follow the Control Plans in accordance with IATF 16949:2016-10. The Control Plans are referenced in the initial sample test report.

5 Defects

5.1 General

In accordance with state-of-the-art technology, the products have to be free from defects, which impair

- the processability under real production conditions
- the use under real service conditions.

Details may be agreed relating to the component.

5.2 Macroscopic internal defects

Macroscopic internal defects are e.g.

- non-metallic inclusions with a surface of more than 0,03 mm² in their largest dimension
- internal discontinuities as e.g. blowholes, pores or internal cracks with a surface of more than 0,03 mm² in the largest dimension

Quantity and size of macroscopic internal defects must not exceed the current state-of-the-art technology. State-of-the-art measures are to be taken in order to avoid the occurrence of macroscopic internal defects. Material with a heightened risk of such defects must not be delivered. In particular, any deviation from the control plan can heighten the risk.

6 Chemical composition (product analysis)

6.1 Deviating from DIN EN 10277:2018-09

Elements not listed below must not be added to the steel intentionally without approval by the orderer, with the exception of the final treatment of the melt. All adequate preparations have to be made in order to avoid the addition of such elements from the scrap and other materials used in manufacturing that impair the mechanical properties and the usability of the steel.

Mass fraction in %		Mass fraction in %	
C	0,04 to 0,10	P	max. 0,15
Si	max. 0,09	S	0,25 to 0,35
Mn	0,99 to 1,26		

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7 Material conditions, types

Bosch designation	Treatment condition	Relevant production sequence
A01 A02	+C+A	cold drawn, heat-treated[1)] to achieve certain magnetic properties
C01	+C	cold drawn (without annealing treatment)

[1)] The physical and mechanical values are strongly influenced by the annealing treatment.

8 Surface qualities

8.1 Surface B3

Smooth, free from loose surface scale, surface imperfections as per DIN EN ISO 8785:1999-10 must not exceed state-of-the-art technology.

Surface quality class 2 as per DIN EN 10277-1:2018-09 must be met.

8.2 Surface B4

Smooth, clean metallic surface, protected with suitable antirust agent. Surface imperfections in accordance with DIN EN ISO 8785:1999-10 must not exceed the state-of-the-art technology.

Surface quality class 2 as per DIN EN 10277-1:2018-09 must be met.

9 Mechanical properties

9.1 Round bars as per DIN EN 10277:2018-09

Delivery condition	Nominal thickness	Tensile strength	Yield strength	Elongation after fracture A	Hardness
	mm	MPa	MPa	%	HBW 5/750
C01Z	from 5 to 10	510 to 810	min. 440	min. 6	165 to 240
	over 10 to 16	490 to 760	min. 410	min. 7	150 to 225
	over 16 to 40	460 to 710	min. 375	min. 8	135 to 210
	over 40 to 63	400 to 650	min. 305	min. 9	120 to195

9.2 Round bars after the annealing treatment deviating from DIN EN 10277:2018-09

Delivery condition	Nominal thickness	Tensile strength
	mm	MPa
A01Z A02Z	from 5 to 10	min. 300
	over 10 to 16	min. 300
	over 16 to 40	min. 300
	over 40 to 63	min. 300

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10 Additional requirements

10.1 Microstructure (in delivery condition)

Ferrite and lamellar pearlite, no ferrite/pearlite bands, which may impair the machining.

10.2 Straightness

The straightness has to guarantee a flawless processability, permissible deviation from straightness max. 1,0 mm/m according to table 4 in DIN EN 10278:1999-12.

10.3 Microscopic degree of purity

10.3.1

The manganese sulfides shall be distributed as homogeneously as possible over the cross-section. Size and distribution of the manganese sulfides have to correspond to standard diagram plate for Stahl-Eisen Testing Guideline SEP 1572:1971-08, column 2, lines 2 and 3.

10.3.2

The manganese sulfides shall be distributed as homogeneously as possible over the cross-section. Manganese sulfide size and manganese sulfide distribution have to correspond to standard diagram plate for Stahl-Eisen Testing Guideline SEP 1572:1971-08, column 1, lines 1 and 2 and column 2, lines 1 and 2.

10.4 Coercive field strength of the annealed round bars

The heat treatment to achieve magnetic properties is performed in accordance with an internal steel mill process specification that has to be coordinated with the responsible CoC Materials. This process specification is part of the control plan, which is referenced in the initial sample test report.

Nominal thickness	Coercive field strength
mm	A/m
up to below 14	max. 300
from 14 to 20	max. 250
over 20	max. 200

11 Certificate of materials testing

An inspection certificate 3.1 in accordance with DIN EN 10204 (2005-01) needs to be provided for each delivery.

12 Additional agreements

Additional details with reference to product geometry, identification, form of delivery and packaging can be stipulated when ordering.

13 Suppliers

No agreements.

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14 Declaration

The supplier accepts the obligation of compliance with the Bosch-Norm N 2580-1 (Prohibition and declaration of substances). The current version should always be used for the declaration (e.g. in the download section of the Bosch Internet). The moment of signing the draft of the order specification is the latest point in time for making the declaration.

15 Density (reference value for the demand analysis)

7,85 g/cm^3

16 Normative references

The following documents are applicable for using this Bosch-Norm. With dated references, solely the edition referred to applies. With undated references, the last edition of the indicated document applies (including all revisions). Intentionally cited historical documents are additionally identified as "H" in the column "Type."

No.	Document number (edition date)	Title	Type
1	N23A 2E01	Material dimensional deviations; Long products	M
2	N 2580-1	Prohibition and declaration of substances	M

Type M = document to be supplied; Z = cited document; H = historical document

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